Wireless local loop

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Wireless local loop systems are available to provide both broadband and narrowband services to customers. Narrowband systems are generally either based on cordless or mobile standards, or a manufacturer's proprietary technology. They are generally designed to offer voice-band services although some are capable of delivering ISDN. The broadband systems are either circuit or cell based. These systems usually operate at millimetre-wave frequencies where a line-of-sight path is required between the customer and a central base-station. Services are typically leased line, ATM and LAN interconnect.

1. Introduction

Over the last two years, wireless has made a significant impact on the local loop both by changing the economics of providing fixed services and by enabling limited mobility about a fixed network terminating point. This paper will focus on the technologies for implementing a wireless local loop (WLL) by replacing all or part of the local access network with a wireless link to a fixed customer radio unit. The use of cordless telephone technologies and of radio LANs to provide limited mobility (local area mobility) in the vicinity of fixed access termination points has been covered in a previous paper [1].

Point-to-point microwave systems have been used for a number of years to deploy greater than 2 Mbit/s services either where fixed access is difficult to provide or to enable rapid deployment. Each customer served by such a system has a dedicated installation at both ends of the link.

All of the new midband and broadband systems (bearer rates greater than ISDN2) and narrowband systems (POTS to ISDN2) are point-to-multipoint, i.e. a group of customers receive their service from the same base-station. Narrowband point-to-multipoint systems are primarily being used for residential applications whereas the broadband systems are targeted at small-to-medium businesses.

This review provides an introduction to the technology of point-to-multipoint radio systems. Only digital systems will be discussed. There will be no mention of the mature proprietary analogue products being used both for some rural applications in developed markets and for more general applications in developing markets.

The key features and attributes of the various options will be stressed. The choice of which systems to deploy in a given situation is dependent upon the regulatory situation, legacy platforms and commercial issues, such as target services and volumes, costs and potential revenues. These factors will not be discussed here as they vary from country to country and from operator to operator.

2. Spectrum

Spectrum is a scarce resource and this limits the allocation available for wireless local loop applications, especially at those frequencies below 4 GHz which are most suited to long-range systems.

Spectrum allocations made for narrowband wireless local loop applications vary from country to country but in most cases they are in the 2 to 4 GHz band. Individual operators have been granted spectrum allocations of approximately 50 MHz within this band. In addition, some cellular operators are considering using their spectrum to support both mobile and fixed services. There is also interest in operating fixed links using the technology and spectrum designated for residential and business cordless use, e.g. digital enhanced cordless telecommunications (DECT) and radio LANs.

Broadband systems require more spectrum and the allocations which have been made, or are being considered, lie within the 10 to 40 GHz band. Figure 1 indicates the current and likely allocations in a number of countries.

ETSI have a proposal known as the broadband radio access networks (BRANs) project which is intended to develop a set of standards for both local area mobility and fixed network access. The system that supports mobility is an extension of the HiPerLAN standards (to be known as HiPerLAN2) which will operate in the 5 GHz band. No spectrum has yet been allocated for the fixed access

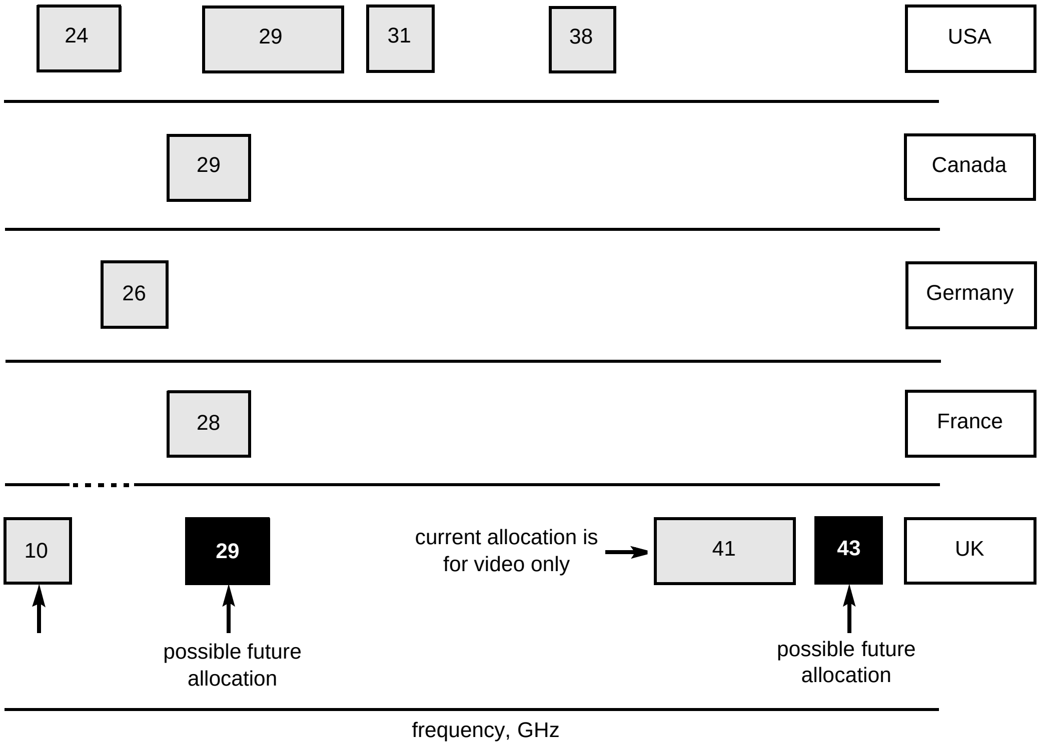

Fig 1 Present distribution of spectrum for broadband point-to-multipoint system.

network system (to be known as HiPerAccess) but spectrum around 40 GHz is a likely candidate.

3. Attributes of narrowband systems

3.1 Architecture and cost structure

The functionality of the elements of a narrowband wireless access system is outlined below.

- Customer radio unit (CRU)

  The CRU will usually be externally mounted, having a wired feed to an internal interface/multiplex unit, power supply and battery back up. Each customer can have their own CRU or a multiple line unit can be shared between customers, i.e. service would be provided to a radio unit mounted on the top of a multi-occupancy office block, with copper feeds from the interface/multiplexer unit to individual customers. For long-range applications, the CRU will have a directional antenna pointed at the base-station. If a flat-plate antenna is used it can be an integral part of the box housing the radio unit.

- Radio base-station

  The link between the radio base-station and the controller can be HDSL, fibre, point-to-point radio, or broadband point-to-multipoint. The base-station can either have an omni-directional antenna or else the capacity can be increased by deploying sectored antennas.

- Radio controller

  The controller provides the match to the interface of the local exchange. Depending on this network, the interface can either be non-concentrating, e.g. V5.1 or DASS2, or concentrating, e.g. V5.2.

- Operational and maintenance (O&M) system

  All the systems on offer have O&M systems which can monitor performance down to individual customer radio units.

Two of the key attributes of WLL are:

- that the cost of the CRU is not incurred until service is offered,

- the cost per home passed of the shared infrastructure can be very low compared with wired technologies.

However, it is necessary to operate at maximum fill to bring the cost per customer served down to competitive levels. This can be a problem for some of the smaller cell systems, although the impact of fill and cell size can be partly offset if there is a high degree of modularity in the base-station build. Manufacturers' offerings differ considerably in the amount of modularity provided and thus the choice of system must be carefully matched to the target market.

The other key attributes are rapid deployment and the ability to reuse (redeploy) equipment.

3.2 The radio channel

The performance of a digital radio system is expressed in terms of the bit error rate (BER). The BER is related to the signal-to-noise ratio at the receiver, where the noise is the sum of thermal noise added by the receiver, interference from other carriers, inter-symbol interference arising from both multipath propagation and filters, etc. In any radio system the choice of modulation method, coding and multiple access method is governed by the need for an adequate signal-to-noise ratio under the conditions imposed by the radio transmission path.

Modulation is the process by which the message is added to the radio carrier. In some systems the modulation method is fixed but in others it is changed to suit the required bit rate.

Channel coding is employed for the detection and correction of errors caused by noise or interference. Transmission coding matches the signal to the characteristics of the channel. Security coding (encryption) prevents unauthorised access to the signal.

Multiple access schemes

Various multiple access (multiplexing) techniques are used to combine individual channels in the radio spectrum to allow for simultaneous access by a number of users, namely:

- frequency division multiple access (FDMA),
- time division multiple access (TDMA),
- spread spectrum — direct sequence code division multiple access (CDMA) and slow frequency hopping (fast frequency hopping is not used by any of the present systems).

Specific radio channels are assigned to individual users by one of two methods. With fixed assignment the customer always has the same radio channel, i.e. has a dedicated link, whereas with demand assignment the customer is allocated a free channel when a call is made. One of the advantages of dynamic channel assignment is that the allocated channel can be changed during a call if the quality of service is being affected by interference.

Transmission delay

The way in which the multiple access is implemented, the channel coding and the codec will all affect the transmission delay. The two-way transmission delay for some systems (e.g. GSM) can be appreciably greater than for wireline systems.

Frequency planning and capacity

Depending on which multiple access method and channel assignment method is chosen there may be a need for frequency planning to reduce interference between base-stations (and customer units) in different cells. When frequency planning is required, each cell would have a different frequency from its neighbour.

The number of frequency sets needed to provide ubiquitous coverage depends on the frequency reuse pattern which needs to be used with a given type of system. It also depends upon the frequency used. Higher frequencies are more severely attenuated by obstacles and thus a tighter frequency reuse pattern can be used. The reuse pattern can also be influenced by using antenna down-tilt at the base-station.

A number of other techniques can be employed to reduce the number of frequency sets needed to provide coverage from overlapping cells. For example, an ‘active’ base-station antenna, i.e. flat-plate antenna whose beam can be steered electronically, can be used to focus on a customer radio unit and thereby reduce interference between cells.

To increase the number of channels which can be provided by a given base-station, a cell can be split up into sectors by using directional antennas at the base-station. In this case the different frequencies will be used in adjacent sectors.

Given that the spectrum allocation will be limited, the frequency reuse pattern will affect the number of channels which can be provided in each cell or sector. Systems can be compared in terms of their spectral efficiency:

$$\text{Spectral efficiency} = \frac{\text{max number of channels per cell * bit rate per channel}}{\text{spectrum allocation required}}$$

A high spectral efficiency is a key differentiator in any competition for spectrum licences and is, of course, a key factor if spectrum has to be bought. It is also a key indicator of the ability to future-proof the system, i.e. to have capacity in hand to meet future needs, especially data.

Some systems use multiple access methods which avoid the need for frequency planning. This significantly eases deployment and can increase spectral efficiency. The main examples are:

- DECT uses dynamic channel assignment — each of the ten frequencies (supporting a total of 120 time-slots) are available in each cell, with interference between cells setting a limit on the number of channels which can be used in a cell at a given time (this limit can be increased by synchronising the transmit/receive cycles of each base-station),

- CDMA replaces frequency planning by code planning,
- slow frequency hopping — the number of hopping sequences will be limited in a WLL implementation to keep interference tolerable, but with radio LANs there is no control of hopping sequences and interference sets the limit to the number of simultaneous users.

3.3 Range

The key factor determining range is the acceptable path loss, which is the difference between the effective radiated power of the transmitter and the receiver sensitivity (threshold), i.e. the minimum received power for an acceptable BER.

The receiver sensitivity will be a decreasing function of channel (bearer) bit rate. Thus low bearer bit rate systems (e.g. GSM) will have a lower receiver threshold than a high bearer bit rate system and, all other things being equal, will have the greatest range.

The effective radiated power (ERP) will depend on the gain of the antenna and the transmit power. The antenna gain can be increased by using directional antennas. The transmit power will be limited by cost, frequency reuse considerations, or regulation. In general, systems can be split into low-power systems (for which the ERP is set by regulation) and high-power systems (for which the ERP is set by the manufacturer). The range of low-power systems, such as DECT, will usually be smaller than that of high power systems, but, as will be seen below, range can also be governed by the environment in which the system operates.

The acceptable path loss has two components:

- an allowance for slow and fast fading,
- a distance and clutter-dependent propagation loss.

Slow fading will depend on factors, such as foliation and rain, which attenuate the radio waves. It is also caused by atmospheric changes, such as temperature inversions, which open up new paths between the transmitter and receiver. These multiple propagation paths cause interference and thereby affect the amplitude of the received signal. The allowance which must be made for slow fading will depend on frequency, the average link length and the environment in which the system is deployed.

Fast fading is caused by reflections or diffractions from objects which are changing positions, such as vehicles or even branches swaying in the wind. The effects of fast fading can be reduced by using highly directional antennas and implementing diversity.

Subtracting the fading allowance from the acceptable path loss gives the allowance for propagation loss and it is this element which determines the range of the system (cell radius). The relationship between propagation loss and distance is influenced by antenna height, environment and frequency.

For the frequency range 3 to 6 GHz the path needs to be almost line of sight. Thus the cell planning tool needs to take terrain and buildings into account. Such tools exist, although databases for urban areas can be expensive. For a given base-station location, such a tool will typically code the surrounding area either red, amber or green. Green means that service can be provided with no problems, amber needs an on-site assessment and red indicates that service is not possible. The base-station antenna and the customer radio units will both need to be above roof tops, and a regular array of base-stations will have to be deployed to increase the probability of a customer being able to get coverage from at least one base-station. Within UK urban areas it appears that a 6 km grid will make it possible to serve most properties.

Below 3 GHz, obstacles such as buildings and terrain features have less effect on range. There are a number of planning tools developed for cellular systems which can be used. However, the acceptable path loss for WLL systems is, generally speaking, significantly smaller than for most cellular systems and the prediction error of the planning tools is thus more significant.

The planning tool error reduces as the customer radio unit is raised above the local clutter. When the CRU is mounted at eaves height the planning tool will typically have an rms error value of 10 to 12 dB. With such mounting the range will be about 4 km in the least dense urban areas and about 2 km in the most dense urban areas.

For low-power WLL systems the planning tools are less effective because a single obstacle on the path can account for most of the allowed path loss. A number of advanced propagation tools have been developed for such small cell systems, but they have not yet proved to be effective even when the environment can be accurately modelled. Thus for low-power systems such as DECT it is necessary to rely on simple planning guidelines which will depend on the type of area in which it is deployed. With eaves-mounted customer radio units, the range will typically be about 2 km in the least dense urban areas and about 1 km in the most dense urban areas, but in certain applications the range can be appreciably less. For example, if the base station antenna is below roof-top height, the range can be less than 100 m.

Below about 1.5 GHz, obstacles and terrain have much less effect on the path loss and longer ranges are possible. Cellular planning tools have been optimised for these frequencies. Within developed markets it is unlikely that this spectrum band will be made available for WLL.

No matter what the operating frequency, if the decision whether or not to provide service to a customer is based on measurements at the customer's premises, the criterion used must include a margin for slow and fast fading. It is not sufficient to show that performance of the system is acceptable on the day of installation.

Reflections will cause multiple propagation paths, with different propagation delays. The combination of these at the receiver causes dispersion, i.e. the spreading of pulses. This results in inter-symbol interference and hence a degradation in the signal-to-noise ratio.

Dispersion can be limited by using directional antennas at both ends of the link, but in some systems it is also necessary to implement correction at the receiver (e.g. equalisation) to prevent the range being limited by dispersion rather than path loss.

3.4 Speech and data capacity

With most point-to-multipoint systems, the radio link provides a concentrating interface. It is thus necessary to use traffic theory to calculate the number of customers who can be supported by a given sector.

The number of channels per sector claimed for a particular technology needs critical assessment. In some systems (e.g. DECT and CDMA) the capacity can be either interference limited or affected by multipath propagation.

The concentrating interface of radio access helps to reduce the cost of providing speech services. However, this advantage could be lost when long-session Internet traffic is carried. To handle such traffic efficiently the radio system needs to detect when it is being used for data and then release the bearer during idle periods, i.e. move to a quasi-packet mode. Manufacturers are planning to implement ways of efficiently handling packet data, but details are not yet available.

There are a number of non-concentrating radio systems. These have, in general, been produced for remote rural applications. They are not considered here.

4. Review of current narrowband systems

4.1 Low-power systems

The key systems within this category are based on the DECT, PHS and PACS cordless standards. In these systems the power is limited to avoid interference with domestic and office products. There is a possibility that transmit powers could be raised if the additional spectrum was found for such WLL systems. The possibility of extending the DECT band to allow WLL applications has been discussed by some of the regulatory bodies in Europe, but no decisions have yet been made.

The decision whether to allow DECT spectrum to be used for WLL purposes is still being argued on a country-by-country basis and many of the present trials are running on special licences. It should also be noted that, although the PHS lobby is claiming that PHS and DECT can share the same spectrum, this is not yet generally accepted.

The present systems offer a 32 kbit/s ADPCM speech channel. The codecs used limit data to 9.6 kbit/s but DECT standards are in place which will provide flexible data pipes. The first of the new DECT chip sets conforming to these standards will be compatible with 28.8 kbit/s modems; these will be followed by chip sets offering 32 and 64 kbit/s interfaces. PHS manufacturers are indicating that they will match this performance.

The key difference between the systems offered by the various manufacturers is their modularity, i.e. the maximum number of carriers per sector, and the number of base-stations which can be supported by a radio controller. Thus a given manufacturer's system may be particularly suitable for serving a specific target market.

Some manufacturers can provide repeaters which can be used to fill in coverage blackspots. They can also be used to provide localised areas of coverage to handsets (typically with a range of less than 100 m) inside or outside buildings.

DECT WLL products are available from Ericsson, Siemens, Alcatel and Lucent, while NTT have a PHS-based WLL system. When advocates of PHS and PACS compare these systems with DECT, much is made of the fact that DECT has a higher transmission rate and that this makes the receivers less sensitive, and therefore more susceptible to dispersion caused by multipath propagation. However, the effects on performance of DECT is not significant in most WLL applications. (The difference in receiver sensitivity will change the cell radius by at most 10% and the effect of multipath propagation is insignificant when directional antennas are used.)

At present most of the manufacturers only make single line CRUs but all have plans to produce multiline units which can be shared between a number of customers.

The only other systems at present in low-power category are some low-capacity CT2 systems intended for niche applications and the Tadiran system. This uses slow frequency hopping and has a 32 kbit/s ADPCM speech channel. The system deployed in the UK (by Atlantic Telecom) has a low transmit power because it operates in the industrial, scientific and medical band (2.4 GHz). Higher power versions are believed to have been developed for deployment in other frequency bands.

4.2 High-power systems

These comprise both a number of systems based on cellular standards, and a wide variety of proprietary systems.

Cellular

The possibility of supporting both fixed and mobile services from the same cellular network infrastructure (e.g. GSM or DCS1800) is being pursued by several mobile operators as a means of providing fixed/mobile integration. An attractive range of fixed customer units for cellular networks are now on the market. The key issues are:

- the relative tariffs of fixed and mobile services which use the same infrastructure,
- whether the low bearer rate services are too limited and inflexible to meet future demand.

Cellular-based fixed customer units are also being deployed expediently by some fixed network operators to provide temporary service when there is an unacceptable delay in providing or repairing wired access.

WLL products based on stripped down cellular systems, i.e. without mobility, are being offered by several manufacturers. There are analogue and digital systems, but only the latter will be considered here. They have primarily been produced for developing markets. This is reflected in the range of frequencies at which the systems are configured to operate. In general, these are below 1 GHz, i.e. in bands which are not available for fixed radio access in developed markets.

Proprietary systems

There are some high-power proprietary systems which are being targeted at the developing markets. These are characterised by operating frequencies below 1 GHz (and thus large cell sizes).

Most of the high-power proprietary systems are targeted at urban and suburban deployment in developed markets and, in general, they operate at above 2 GHz. The manufacturers include DSC, Lucent, Nortel and Teledata. There are moves to produce standards based on all of these manufacturers' products, but time-scales are unclear.

At present all the customer radio units are only available for individual customers, but most manufacturers have plans to produce multiline units which can be shared between a number of customers.

Nortel's system is a conventional TDMA system based on 32 kbit/s bearers. It is by far the most mature of the WLL systems and is already in service in several countries. The customer radio units provide three 32 kbit/s bearers. One is used for 32 kbit/s ADPCM speech; the other two can be used for a second speech circuit or to support a 28.8 kbit/s modem. Nortel plan to have an ISDN basic rate access version by the end of 1998.

A number of manufacturers are using CDMA. Lucent (AirLoop) uses 16 kbit/s bearers which can be combined to give 32 and 64 kbit/s channels, etc. DSC's system (Airspan) is based on 64 kbit/s channels and they are at present providing customer units which offer two PSTN lines. Siemens (CDMA link) and Krone (Telecell M) are also developing CDMA systems.

Teledata have an FDMA system which adapts the modulation method to match the data rate. Their customer radio units support either:

- 4 lines of ADPCM at 32 kbit/s,
- 2 lines of PCM at 64 kbit/s,
- 2 lines of digital data at 64 kbit/s,
- 1 line of ISDN basic rate access.

One merit of the Teledata system is its simplicity.

5. Midband and broadband systems

A number of midband systems capable of delivering multiple ISDN2 lines have been developed from the narrowband wireless local loop products. They are primarily for the delivery of circuit-switched services and are intended to provide symmetrical access. The systems offer capacity of typically 16 Mbit/s per cell and are most suitable for delivering service of under 1 Mbit/s. Typical applications are delivery of leased line and ISDN services to small businesses in urban and suburban areas. They can also be used to provide backhaul to cellular network micro-base-stations. The key attributes of these systems are discussed in more detail in section 6.

Broadband systems have been developed to deliver a more flexible range of services. Initially the development of these systems was focused on microwave video distribution systems (MVDS). The early versions of these systems were one way and were intended for residential broadcast services but two-way systems are in development for interactive multimedia services (IMS) and IP applications. Most attention is now focused on symmetrical and asymmetrical local multipoint distribution systems (LMDS) which support ATM. Typical applications are the provision of multiple ISDN lines and flexible data interfaces to small-to-medium businesses in urban and suburban areas. The LMDS developments have been driven by the USA market, with the unique selling point being that the customers can easily reconfigure their terminals to upgrade their services. The key attributes of the LMDS systems and the present state of the market is discussed in section 7.

Both LMDS and MVDS are based on the MPEG frame structure and it is possible that hybrid systems may emerge which support both residential IMS and business services from the same hub.

The ETSI broadband radio access networks (BRAN) project is taking a longer term view to broadband wireless access. It is intended to provide a set of functional specifications by January 2000. User bit rates are intended to be from 16 kbit/s to 16 Mbit/s, and systems should support coexistence and interoperability in the physical layer. Since the standards are still in development, they will not be discussed in any more detail in this paper.

6. Midband systems

6.1 Architecture and costs

The basic architecture of midband systems is very similar to that of narrowband wireless local loop. The customer installation will typically consist of an outdoor radio unit and an indoor multiplexing and interface unit for connection to the customer. At present interfaces are typically a mix of POTS, ISDN2 or E1 depending on the line cards installed in the CRU. The E1 can be delivered as a fractional service to provide sub-2-Mbit/s bit rates. The bit rate allocated to a customer can be dynamically allocated to take advantage of the concentrating radio interface.

The radio base-station will typically operate in a four-sector configuration to allow better use of limited spectrum. Backhaul from the base-station to the radio controller will typically be using either fibre or point-to-point radio.

The radio controller will control dynamic bandwidth allocation, as well as providing interfaces to the local exchange and management system. Interfaces will typically be based on V5 protocols with Q.931 being used for ISDN services.

CRUs for these systems typically cost around £3k and base stations are around £40k for a 16 Mbit/s capacity.

6.2 The radio channel

All of the systems currently available for this market (such as Floware/Siemens WALKair, Alcatel 9800HC and Siae PMP60) use TDMA. They are designed to operate in either the 3.5 GHz or 10.5 GHz bands.

Deployment at 3.5 GHz is very similar to narrowband wireless local loop systems operating in the same frequency band. As a near line-of-sight path is required, planning tools are needed that model propagation using a range of inputs based on local terrain and buildings.

At 10.5 GHz a line-of-sight path is required between the customer and base-station. Therefore deployment is very dependent on local building topology. The problems of planning such systems are considered in section 7.2.

6.3 Services and capacity

All the systems are currently intended to deliver circuit-switched applications. Most of the systems available will typically be able to offer up to eight POTS lines per customer, four ISDN2 lines per customer or one E1 connection. As services can be mixed, a customer could, for example, be provided with four POTS lines for a small PBX, an ISDN port for dial-up data and a 256 kbit/s leased line.

Typically the maximum throughput per customer unit is 2 Mbit/s and the number of line cards that can be supported is limited. For some systems the number of 64 kbit/s channels allocated on the E1 must be configured via a management system, whereas other systems can respond dynamically to customer demand.

At present none of the systems can offer a packet-based service without using an external multiplexer on an E1 connection, but it is expected that second generation systems will be optimised to support packet-based data connectivity.

The base-station is typically deployed using four sectors, each offering 4 Mbit/s. This means that a system could be deployed in a 2 × 14 MHz spectrum allocation which would fit within a narrowband allocation at 3.5 GHz. Therefore these systems allow operators to deploy a product that can use existing narrowband spectrum but offer services to small businesses.

7. Local microwave distribution system

7.1 Architecture and costs

The basic architecture of an LMDS system is illustrated in Fig 2.

The customer unit comprises a roof-based antenna and RF unit which typically has a 1 GHz intermediate frequency feed to an indoor unit which can accept a number of interface cards. This unit can serve multiple customers.

The range of customer interfaces which will be needed to provide the most attractive market offering are still being developed. The initial USA product offerings will be based on T1 and 10BaseT, but E1-based options are being developed for Europe. The next generation is likely to include unstructured T1 and E1 and 25 Mbit/s ATM UNI.

For the majority of the broadband systems the RF link employs cell-based protocols. There is no cell transmission during idle periods and thus the concentrating radio interface is optimised for bursty data.

At the hub there is a roof-top mounted RF unit which can have sectored antennas. Most of the early deployment will be based on up to four sectors per hub, but it is feasible to evolve to 24. At LMDS frequencies antenna polarisation can be used to reduce interference between sectors in a cell and between cells, i.e. if one sector is vertically polarised, horizontal polarisation will be used for neighbours that are likely to cause interference. In principle, the same frequencies could be used in all sectors (i.e. a frequency reuse of one), but it is possible that operational experience may support the view that two frequency sets should be used.

There will be baseband feeds to an indoor controller which will have SDH or Sonet interfaces to an ATM network which will provide distribution and break-out for the various services supported.

The cost of customer units is typically £4k at present but this could soon fall to £2k as volumes increase. For a fully loaded base-station the shared infrastructure cost can be of the order of £4k per customer unit supported. However, in some deployment scenarios the equipment cost could be less significant that the rentals and leases for the roof rights in urban and suburban areas.

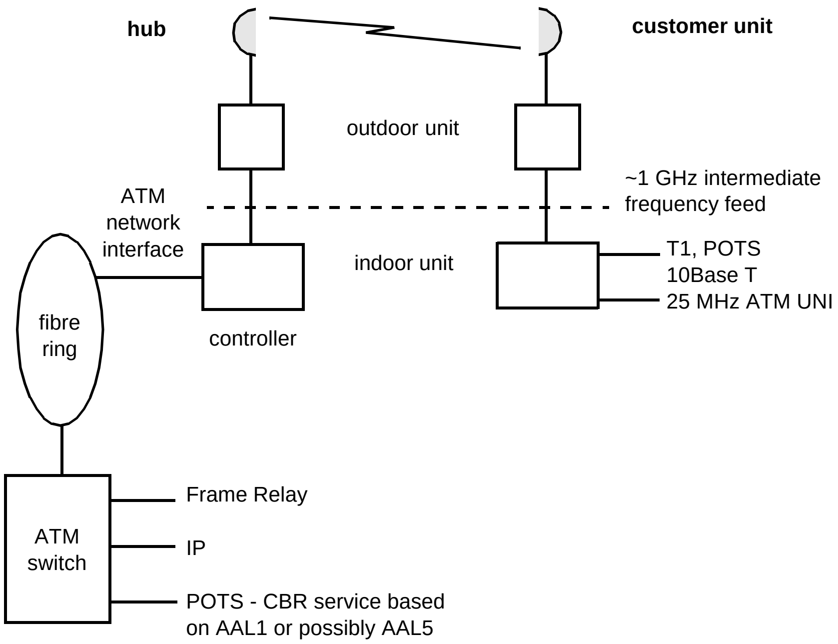

Fig 2 Main elements of an LMDS system.

7.2 Spectrum requirements

In most systems the upstream and downstream carrier bandwidths are variable and can be optimised for both customer needs and spectrum allocation. Different frequency bands are required for the downstream and upstream traffic, and in most countries the spectrum allocation is being based on separate tranches for upstream and downstream traffic, rather than on a contiguous band. In general, the closer the bands, the more costly the filtering required.

7.3 Range

At LMDS frequencies all links must be line of sight. Thus shadowing by tall buildings will limit the range and it will not be feasible to reach everybody in a cell. Planning requires site surveys at present, but innovative techniques based on aerial photography are emerging. Surveys in the UK indicate that in town areas it is reasonable to expect to cover 70% of the buildings within a 2 km radius of a roof-mounted base-stations. With a regular array of such base-stations, this probability can rise towards 90%.

At frequencies above ~20 GHz rain fades can significantly limit the range and it is thus necessary to plan for the occasional very heavy rainfall which can cause a complete link outage. Comprehensive rain fall statistics exist and using them it is possible to plan for the 99.999% availability needed to be able to market a ‘fibre quality’ service.

In many countries the need to be able to cope with rain fades has an impact on the modulation methods which can be employed. Higher order modulation methods increase the bit rate which can be supported by a given carrier bandwidth, for example:

- QPSK 1.5 bit/s per Hz (including guard bands),
- 16 QAM 3.5 bit/s per Hz (including guard bands),
- 64 QAM 5 bit/s per Hz (including guard bands).

However, because higher order modulation methods require a larger signal-to-noise ratio, it can be necessary to drop back to QPSK during rain fades in order to maintain a given link length. During such fades priority would be given to delay-sensitive services such as voice, but there is debate within the industry as to whether such protocols would be acceptable to the customer.

In summary, the choice of modulation method thus depends on the rainfall region in which an LMDS is to be operated, the amount of bandwidth available, the line-of-sight link length in the target geotypes, and the service variations which the market will accept.

7.4 The radio channel

The downstream (hub to customer unit) and upstream (customer unit to hub) channels must be considered separately.

Downstream

All manufacturers’ systems support multiple carriers per sectors. In the initial implementations it is likely that a customer unit will only access a single carrier, but the possibility of multicarrier customer units is being discussed.

The number of carriers and the bandwidth per carrier vary from manufacturer to manufacturer, but in most cases the bandwidth can be varied to suit customer needs and the spectrum allocation.

Modulation is typically QPSK which gives a spectral efficiency of around 1.5 bit/s per Hz and time division multiplexing (TDM) is used on each carrier.

Early deployments are likely to be able to provide approximately 250 Mbit/s per sector. Depending on spectrum allocations, capacities of greater than 750 Mbit/s per sector are feasible.

Upstream

In order to reduce costs, the hub transmitter power per carrier is usually greater than that for the customer unit. Given this difference, the upstream link budget can only be made comparable with the downstream link budget if:

- the customer unit can only transmit on a single carrier,
- the bandwidth of the carrier is lower than that for the downstream channel.

Consequently the upstream capacity of a customer unit is lower than its downstream capacity (typically 12 Mbit/s and 50 Mbit/s respectively).

The bandwidth of the upstream carrier is usually flexible and in some implementations it can even be changed dynamically to match traffic conditions.

The first generation of customer units has a dedicated upstream carrier and in some implementations the bandwidth allocated to the dedicated carrier can be varied in response to the user’s upstream demands. Such a system makes very efficient use of spectrum but the frequency management protocols can be complex.

Second generation customer units will use TDMA to share the same carrier. This option offers greater flexibility and makes it easier to offer dynamic bandwidth allocation, but it increases the complexity of the media access control (MAC).

7.5 Capacity sharing issues

The capacity of LMDS is more than adequate for all the likely speech applications based on $n \times 64$ kbit/s bearers and a flexible range of data services can be added at marginal extra cost. It must, however, be stressed that the downstream capacity is shared by all customers and that the upstream capacity per customer unit is limited (to typically 12 Mbit/s at present). The upstream capacity can be enhanced by adding extra radio channels at the customer unit and by technology advances (e.g. a reduction in the cost per W of transmitters), but the consequences of capacity sharing and a limited upstream capacity needs to be considered when evolution planning.

7.6 Current status

A number of LMDS systems are now coming to market, but it is too early at this stage to give a detailed breakdown of their differences. In some cases there has been a merging of companies, e.g. Texas Instruments/Bosch, BNI-Nortel, to capitalise on their individual strengths across the areas of RF, baseband and network management. Other traditional telecommunications suppliers with extensive RF expertise, e.g. Alcatel and HP, are using OEM supplies of baseband LMDS technology, e.g. Stanford Telecom. All aim to supply turn-key systems with comprehensive network management responsible for billing data, customer management, and performance and alarms.

8. Conclusions

The key attributes of narrowband wireless access are that the service area can be expanded rapidly and that:

- the cost per km^2, and the cost per home passed, of the shared infrastructure is significantly lower than for wired access,

- the cost of the customer radio unit need only be incurred when service is provided — this cost is the largest element in the cost per customer but it should fall significantly as volumes increase,

- the operating costs of wireless access should be significantly lower than for wired access if there are no reliability problems and the churn is low,

- multiple lines per customer can be provided at marginal extra cost,

- as the present generation of wireless access systems could have difficulties coping with data traffic, and this may have a significant effect on peak traffic rates, most manufacturers are intending to implement procedures which will go some way to alleviating this potential problem.

Despite these attributes, the cost per customer connected is unlikely to be significantly lower than the wired access of an established operator. Thus the best policy for established operators will be to capitalise upon their existing infrastructure rather than invest in wireless access. Wireless access will, however, be an attractive option for niche applications such as greenfield sites and for services to locations such as tower blocks, where maintenance costs for wired access are high.

For new entrant operators wireless access will provide a low investment route for rapidly rolling out a flexible service package. The key to success will be the degree to which the service package can be marketed and bundled to attract high revenue customers and to lock them in. Low revenue customers and/or high churn would be a disaster. Bundling could involve providing a range of data services or combining a fixed service with a cellular or a local area mobility package. In countries where the ISDN2 has been rolled out to most customers, the attractions of wireless access are likely to be less because these customers already have a flexible access bearer. In this case a new entrant operator would have to compete on price only. Given that the capital costs of a wireless solution are comparable or higher than that of the established operator, the ability to undercut tariffs will depend on achieving low operating costs and this could be undermined by poor reliability (especially of customer radio units) or high churn.

The key attribute of broadband radio access is that it is possible to provide multiple PSTN or ISDN lines to small-to-medium businesses and to offer a flexible data service at marginal extra costs.

Reference

1 Merrett R P and Buttery S: 'Cordless technology in a mobile environment', BT Technol J, 14, No 3, pp 55—63 (July 1996).

Bob Merrett joined BT as an apprentice in 1965. After graduating from the University of Wales with a degree in Electrical and Electronic Engineering in 1969 and a PhD from the University of Cambridge in 1972, he joined the reliability division at BT Laboratories where he led teams assessing the reliability of semiconductor packaging, reed relays and a range of discrete electronic devices. In 1983 he joined the devices division and led a team designing high speed GaAs integrated circuits and then a team developing InP OEICs. Since 1990 he has worked on various aspects of radio local loop and currently co-ordinates the technical projects on wireless access and local area mobility.

Paul Beastall joined BT in 1991 as a sponsored student. As a student he spent six months in the fibre systems unit designing a spread spectrum system to perform ranging on a passive optical network.

After graduation in 1995 he joined the copper, radio and satellite systems unit and has worked on a wide variety of fixed access, cordless and fibre-based projects. His current interests lie with fixed wireless access systems and he provides a consultancy role to a number of BT's overseas operations.

Steve Buttery joined BT in 1981 as an apprentice. He worked on installation for two years before leaving for university, from where he graduated in 1989, to join the radio division at BT Laboratories.

Since then he has worked on several radio related projects, ranging from writing software planning tools to supporting the deployment and testing of rural radio access systems. Currently he is the resource manager for the spectrum management and radio team.